KØBENHAVNS UNIVERSITET
UNIVERSITY OF COPENHAGEN

THOR WESSEL LINDBERG

cpr. 270497-1357

har den 10. juli 2020	*has on 10 July 2020*
opnået	*been awarded the degree of*
bachelorgraden i	*Bachelor of Science in*
sundhed og informatik	*Health Informatics*
og titlen	*and the title*

BSc i sundhed og informatik

Bachelor of Science in Health Informatics

Ulla Margrethe Wewer
Dekan/*Dean*

Tina Lewis
Uddannelseschef/*Director of Education*

DET SUNDHEDSVIDENSKABELIGE FAKULTET
FACULTY OF HEALTH AND MEDICAL SCIENCES

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This is a digital diploma. See www.diploma.ku.dk for further information regarding validation

KØBENHAVNS UNIVERSITET
DET SUNDHEDSVIDENSKABELIGE FAKULTET

Thor Wessel Lindberg

Cpr.: 270497-1357

har gennemført bacheloruddannelsen i
sundhed og informatik
10. juli 2020

Oversigt over prøver og bedømmelser side 1 af 3

Følgende resultater er opnået	Resultat 7-trinsskala	Resultat ECTS-skala	ECTS point
1. semester			
Introduktion til Sundhed og Informatik	Bestået		5,0
Statistiske metoder	4	D	10,0
Modellering og Programmering	4	D	15,0
2. semester			
Videnskabs- og teknologifilosofi og Etik	10	B	5,0
Epidemiologiske metoder	4	D	7,5
Sundheds-IT: System udvikling og databaser	12	A	7,5
Indledende medicinsk billedanalyse	10	B	10,0
3. semester			
Humanbiologi, Sundhed og Informatik	7	C	10,0
IT-projektledelse	4	D	7,5
Bioinformatik	7	C	5,0
IT-sikkerhed	10	B	7,5

3. August 2020

Marie Fly Hansen
Sektion for Studieservice

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10. juli 2020

Oversigt over prøver og bedømmelser side 2 af 3

Følgende resultater er opnået	Resultat 7-trinsskala	Resultat ECTS-skala	ECTS point
4. semester			
Sygdomslære for ikke-klinikere	12	A	10,0
Introduktion til organisationsanalyse og sundhedsvæseners struktur og funktion	7	C	10,0
Interaktionsdesign	7	C	7,5
Projekt i menneske-datamaskine interaktion eller it-projektledelse	7	C	2,5
5. semester - Valgfag			
Kvalitative metoder og co-design af selvhjælpsteknologier	7	C	7,5
Individuelt tilrettelagt studieenhed 2.5 ECTS *Visualisering af statistisk analyse med Python*	Bestået		2,5
Applied Programming for Biosciences	Bestået		7,5
Introduction to Programming *Meritoverført IT-Universitetet i København*	10	B	7,5
Introduktion til programmering og databehandling *Meritoverført Danmarks Tekniske Universitet*	10	B	5,0

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10. juli 2020

Oversigt over prøver og bedømmelser side 3 af 3

Følgende resultater er opnået	**Resultat** 7-trinsskala	**Resultat** ECTS-skala	**ECTS** point
6. semester			
Projektophold	Bestået		5,0
Introduktion til systembiologi	7	C	5,0
Bachelorprojekt i sundhed og informatik	10	B	20,0
Tools for caretakers of stroke victims			

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Kompetenceprofil for uddannelsen

Kompetenceprofil
Efter endt uddannelse har bachelorer i sundhed og informatik opnået nedenstående viden, færdigheder og kompetencer.

Viden
- Kan beskrive det danske sundhedsvæsens overordnede organisering og centrale aktører - Har viden om datatyper og håndtering af data, herunder billeder, i sundhedssektoren.
- Har viden om hvordan håndtering af data kan foregå sikkert og processer kvalitetssikres.
- Har viden om modeller for databehandling og udveksling på makro-, meso- og mikroniveau i sundheds- og omsorgssektoren.
- Har viden om forskningsmetoder til at belyse betydningen af og interaktionen mellem brugerne, teknologien og den organisatoriske tilknytning ved udvikling af teknologibaserede løsninger til sundheds- og omsorgsområdet.
- Har kendskab til internationalt anerkendte projektledelses- og evalueringsmodeller, herunder medicinsk teknologivurdering, og deres anvendelse i sundheds- og omsorgssektoren.
- Kender til menneskets krop og funktioner på mikro- og makroskopisk niveau samt de hyppigste og væsentligste sygdomme. Kan anvende medicinsk terminologi relateret til sygdomslære.
- Har viden om basal statistisk og epidemiologisk teori, metode og ræsonnement til anvendelse med relevans for sundhed og informatik.
- Kan fremsøge, gengive, forklare, bedømme og kritisere videnskabelige artikler på sundheds- og informatikområdet.
- Kan analysere og fortolke empirisk datamateriale eller tidligere indsamlede data.
- Kan definere og afgrænse en problemstilling, der relaterer til sundhed og informatik, og argumentere for valg af metode i relation til den valgte problemstilling.
- Kan identificere og forholde sig til videnskabsteoretiske og etiske problemstillinger i sundhedsvidenskabelige og it-faglige projekter.

Færdigheder
- Har viden om relevante kvalitative og kvantitative metoder i forhold til afdækning af behov, og kan udfærdige kravspecifikationer til teknologiske løsninger i sundheds- og omsorgssektoren og evaluere brugervenlighed.
- Kan selvstændigt behandle, analysere og diskutere en problemstilling, der har udgangspunkt i it og anvendelsen i sundhedssektoren eller den industri, der er leverandør til sundhedssektoren.
- Kan forstå og selvstændigt analysere de organisatoriske problemstillinger, som mødet mellem sundhedsvæsen og informationsteknologi afstedkommer.
- Evner både skriftlig og mundtlig akademisk fremstilling.
- Samarbejde og kommunikere effektivt med sundhedspersonale og sundhedsvidenskabelige forskere, såvel som den almene befolkning.

3. August 2020

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10. juli 2020

Kompetencer
- Kan planlægge og styre et mindre it-projekt på en måde, så der løbende sker en systematisk styring af realiseringen af forretningsmæssig nytteværdi i forhold til omkostninger, tidsforbrug og risikoniveau.
- Kan evaluere, planlægge og udføre design og re-design af brugsvenlige IT-systemer. - Kan selvstændigt indgå i fagligt og tværfagligt samarbejde; herunder forholde sig til faglige, etiske og samfundsmæssige perspektiver i forbindelse med udnyttelse af forskningsresultater i sundheds og velfærdsteknologi.
- Kan selvstændigt tilrettelægge et studieforløb for at sikre en vedvarende opdatering af den viden, der er relevant for bacheloren.

3. August 2020

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UNIVERSITY OF COPENHAGEN
FACULTY OF HEALTH AND MEDICAL SCIENCES

Thor Wessel Lindberg
Cpr. : 270497-1357

has completed the Bachelor's programme in
Health Informatics
10 July 2020

Summary of examinations and grades page 1 of 3

The following grades were awarded	Grade 7-point scale	Grade ECTS scale	ECTS credits
1st Semester			
Introductory Course	Passed		5,0
Statistical Methods	4	D	10,0
Modelling and Programming	4	D	15,0
2nd Semester			
Philosophy and Science, Technology and Ethics	10	B	5,0
Introduction to Epidemiology	4	D	7,5
Health Informatics: Systems Development & Databases	12	A	7,5
Introduction to Medical Image Analysis	10	B	10,0
3th Semester			
Human Biology, Health Informatics	7	C	10,0
IT-Project Management	4	D	7,5
Bioinformatics	7	C	5,0
IT Security	10	B	7,5

3 August 2020

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Summary of examinations and grades page 2 of 3

The following grades were awarded	Grade 7-point scale	Grade ECTS scale	ECTS credits
4th Semester			
Human Diseases for Non-Clinicians	12	A	10,0
Introduction to Organization Analysis and the Structure and Function of Health Care Systems	7	C	10,0
Interaction Design	7	C	7,5
Project in Human-Computer Interaction or IT-Project Management	7	C	2,5
5th Semester - Elective Cources			
Qualitative Methods and Co-design of Self-care Technologies	7	C	7,5
Individualised study unit 2.5 ECTS *Visualizing statistical analysis using Python*	Passed		2,5
Applied Programming for Biosciences	Passed		7,5
Introduction to Programming *Credit transferred IT University of Copenhagen*	10	B	7,5
Introduction to programming and data processing *Credit transferred Technical University of Denmark*	10	B	5,0

3 August 2020

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Summary of examinations and grades page 3 of 3

The following grades were awarded	Grade 7-point scale	Grade ECTS scale	ECTS credits
6th Semester			
Internship	Passed		5,0
Introduction to Systems Biology	7	C	5,0
Bachelor Project in Health Informatics *Tools for caretakers of stroke victims*	10	B	20,0

3 August 2020

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10 July 2020

Skills profile for the programme

Upon completion of the programme, a Bachelor of Health Informatics will have acquired the following knowledge, skills and competencies:

Knowledge
- Is able to describe the Danish healthcare sector¿s overall organisation and key actors.
- Has acquired knowledge of data types and handling, including medical images, in the healthcare sector.
- Has acquired knowledge of how data can be handled safely and processes can be quality assured.
- Has acquired knowledge of models for macro-, meso- and micro-level data processing and exchange in the healthcare and social care sectors.
- Has acquired knowledge of the research methods illustrating the meaning of and interaction between users, technology and the organisational affiliation in the development of technology-based solutions for the healthcare and social care areas.
- Has acquired knowledge of internationally recognised project management and assessment models, including medical technology assessment, and their application in the healthcare and social care sectors.
- Has acquired knowledge of the human body and its functions on a micro- and macroscopic level, as well as the most common and most important diseases. Can use medical terminology related to pathology.
- Has acquired knowledge of basic statistics and epidemiological theories, methods and reasoning for application as relevant to health and informatics.
- Is able to search for, reproduce, explain, review and criticise scientific articles in the health and informatics area.
- Is able to analyse and interpret empirical data material or data collected previously.
- Is able to define and delimit a problem related to health and informatics and present a case for the selection of method in relation to the given problem.
- Is able to identify and relate to scientific theory and ethical problems in medical science and IT-related projects.

Skills
- Has acquired knowledge of the relevant qualitative and quantitative methods used to uncover needs, and is able to draw up requirement specifications for technological solutions in the healthcare and social care sectors and evaluate user-friendliness.
- Is able independently to handle, analyse and discuss a problem based on IT and its use in the healthcare sector or the industry supplying the healthcare sector.
- Is able to understand and independently analyse the organisational problems arising from the meeting between the healthcare sector and information technology.
- Is able to make oral and written presentations in academic terms.
- Is able to cooperate and communicate effectively with healthcare personnel and health science researchers as well as the general public.

3 August 2020

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Competencies
- Is able to plan and manage a small-scale IT project in a way that ensures ongoing systematic control of the realisation of business utility relative to cost, time spent and level of risk.
- Is able to evaluate, plan and carry out the design and re-design of user-friendly IT systems.
- Is able independently to take part in discipline-specific and interdisciplinary cooperation, including relating to professional, ethical and social perspectives in connection with the utilisation of research results in health and welfare technology.
- Is able independently to plan a course of study that will ensure continuous updating of the knowledge relevant for the Bachelor's Degree.

3 August 2020

Marie Fly Hansen
Education and Student Services

Diploma Supplement

This Diploma Supplement follows the model developed by the European Commission, Council of Europe and UNESCO/CEPES. The purpose of the supplement is to provide sufficient independent data to improve the international 'transparency' and fair academic and professional recognition of qualifications (diplomas, degrees, certificates etc.). It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It should be free from any value judgements, equivalence statements or suggestions about recognition. Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why.

1. INFORMATION IDENTIFYING THE HOLDER OF THE QUALIFICATION

1.1. Family name(s): Lindberg

1.2. Given name(s): Thor Wessel

1.3. Date of birth: 27 April 1997

1.4. Danish civil registration number: 270497-1357

2. INFORMATION IDENTIFYING THE QUALIFICATION

2.1. Name of qualification and title conferred *(in Danish):* Bachelor i Sunhed og Informatik (BSc Sundhed og Informatik)

Name of qualification and title conferred *(in English)***:** Bachelor of Science in Health Informatics., (BSc Science in Health Informatics)

2.2. Main fields of study: The bachelor studies of Health Informatics are planned to a total of 3 years consisting of subjects from the areas of Health Science, Computer Science, and Technical Science.

2.3. Name and status of awarding institution: Københavns Universitet, University of Copenhagen (officially abbreviated KU) is a state-financed higher education institution, regulated according to the Ministry of Higher Education and Science University Act no. 960 of 14 August, 2014.

2.4. Name and status of institution administering the studies (See 2.3.)**:** Same as above

2.5. Language(s) of instruction/examination: Danish

3. INFORMATION ON THE LEVEL OF THE QUALIFICATION

3.1. Level of qualification: Theoretically, research based tertiary programme, Bachelor Programme

3.2. Official length of programme: 3 years = 180 ECTS credit points ECTS

3.3. Access requirements: Entrance to Bachelor's degree programmes is subject to the regulations contained in order no. 181 of 23 February 2010 issued by the Danish Ministry of Science, Technology and Innovation.

4. INFORMATION ON THE CONTENTS AND RESULTS GAINED

4.1. Mode of study: Full time study

4.2. Programme requirements: The programme is offered by the Faculty of Health and Medical Sciences in collaboration with the Faculty of Science, University of Copenhagen and the Danish University of Technical Sciences.

4.3. Programme details and individual grades/marks/credits obtained: Please refer to the enclosed grade transcript.

4.4. Grading scheme and if applicable grade distribution information: Please refer to the enclosed explanation of the Danish education system and the grading scale.

4.5 Overall classification of the qualification: Not applicable for Danish qualifications.

5. INFORMATION ON THE FUNCTION OF THE QUALIFICATION

5.1. Access to further study: A Bachelor's degree in a given subject area qualifies graduates for entrance to the Master's degree programme in this area, providing that there is a sufficient number of student places available.

5.2. Professional status: Upon graduation, the graduate is awarded the title baccalaur scientific Health Informatics

6. ADDITIONAL INFORMATION

6.1. Additional information: Founded in 1479 by the Danish King Christian I, the University of Copenhagen is Denmark's oldest and largest institution of research and higher education. More than 37,000 students are enrolled in undergraduate and graduate programmes, plus an additional 2,500 PhD students. Staff members number 9,000. The University is divided into six faculties: Theology, Law, Social Sciences, Health and Medical Sciences, Humanities and Science; all situated in the capital of Denmark.

6.2. Further information:

Faculty of Health and Medical Sciences
Blegdamsvej 3B
DK - 2200 København N.

Phone +4535327900
E-mail: email@sund.ku.dk

General information on higher education in Denmark can be obtained from the following two homepages: Ministry of Science, Technology and Innovation: www.vtu.dk, or Danish Rectors Conference: www.rks.dk

7. CERTIFICATION OF THE SUPPLEMENT

7.1. Date: 3 August 2020

7.2. Marie Fly Hansen

7.3. Education and Student Services

UNIVERSITY OF COPENHAGEN

8. INFORMATION ON THE DANISH HIGHER EDUCATION SYSTEM

June 2016

Public higher education institutions in Denmark are regulated by national legislation concerning degree structures, teacher qualifications and examinations. Accreditation in higher education is undergoing transition from programme-based accreditation to institutional accreditation. Programmes and institutions are accredited by national, independent accreditation agencies and the Accreditation Council.

Higher education institutions

Higher education is offered by five types of higher education institutions:

1. Business academies (Erhvervsakademi) offering professionally oriented short cycle and first cycle degree programmes.
2. University Colleges (Professionshøjskole) offering professionally oriented first cycle degree programmes.
3. Maritime Education and Training Institutions offering professionally oriented short cycle and first cycle degree programmes.
4. General and specialised research universities (Universitet) offering first, second and third cycle degree programmes in academic disciplines.
5. University level institutions offering first, second and third cycle degree programmes in subject fields such as architecture, design, music, and fine and performing arts.

Most higher education institutions are regulated by the Ministry of Higher Education and Science (type 1-5).

The Ministry of Culture regulates a number of higher education institutions offering programmes within fine and performing arts (type 5).

Qualification framework

The qualification levels form the basis for the Danish National Qualifications Framework for Higher Education, which is certified in accordance with the overarching Bologna Framework according to the principles adopted by the European Ministers of Higher Education. Danish higher education qualifications at levels 5-8 of the Danish Qualifications Frame-work for Lifelong Learning (NQF) correspond with levels 5-8 of the European Qualifications Framework (EQF).

Admission and progression

General access to higher education in Denmark requires an Upper Secondary School Leaving Certificate or comparable qualifications. Admission to some particular programmes requires entrance examination or submission of a portfolio of artistic work. Holders of an Academy Profession degree can obtain a Professional Bachelor's degree within the same field of study through a top-up programme. Completion of a first cycle degree qualifies students for admission to the second cycle.

Ordinary Higher Education degrees

The Academy Profession degree is awarded after 90-150 ECTS and includes a period of work placement of at least 15 ECTS. The programmes are development-based and combine theoretical studies with a practical approach. Programmes are, among others, offered within Marketing Management, Computer Science and Chemical and Biotechnical Science. The Danish title is field of study followed by the abbreviation AK and the English title is AP Graduate in [field of study].

Overview of degrees in the Danish Higher Education System

Danish higher education institutions use the European Credit Transfer System (ECTS) for measuring study activities. 60 ECTS correspond to one year of full-time study.

Danish qualifications levels	**Ordinary higher education degrees**	**Adult/Continuing higher education degrees**	**Qualifications Framework for the European Higher Education Area – Bologna Framework**	**European/National Qualifications Framework for Lifelong Learning – EQF/NQF**
Academy Profession level	Academy Profession degree (90-150 ECTS)	Academy Profession degree (60 ECTS)	Short cycle	Level 5
Bachelor's level	Professional Bachelor's degree (180-240 ECTS)*	Diploma degree (60 ECTS)	First cycle	Level 6
	Bachelor's degree (within fine arts) (180 ECTS)			
	Bachelor's degree (180 ECTS)			
Master's level	Master's degree (within fine arts) (120-180 ECTS)	Master degree (60-90 ECTS)	Second cycle	Level 7
	Master's degree (120 ECTS)**			
PhD level	PhD degree (180 ECTS)		Third cycle	Level 8

** Can be obtained through a full regular bachelor's programme (180-240 ECTS) or a top up bachelor's programme (90 ECTS) following an Academy Profession degree. A few Professional Bachelor programmes are 270 ECTS.*
*** A few Master's programmes are up to 180 ECTS.*

The Professional Bachelor's degree is awarded after 180-270 ECTS and includes a period of work placement of at least 30 ECTS. The programmes are applied programmes. They are development-based and combine theoretical studies with a practical approach. Examples of professional bachelor's degree holders are nurses, primary and lower secondary school teachers and certain types of engineers. The Danish title is Professionsbachelor i [field of study] and the English title is Bachelor of [field of study].

The Bachelor's degree from a university is awarded after 180 ECTS. The programmes are research-based and are offered in all scientific fields. The Danish title is Bachelor (BA) i [field of study] or Bachelor (BSc) i [field of study] and the English title is Bachelor of Arts (BA) in [field of study] or Bachelor (BSc) of Science in [field of study].

The Bachelor's degree (within fine arts) is awarded after 180 ECTS. The programmes are based on research and artistic research. Programmes are offered within the fine arts. The Danish title is Bachelor (BA) i [field of study], Bachelor i musik (BMus) [field of study] or Bachelor i billedkunst (BFA) [field of study] and the English title is Bachelor of Arts (BA) in [field of study], Bachelor of Music (BMus) [field of study] or Bachelor of Fine Arts (BFA) in [field of study]. A higher education degree within theatre or filmmaking is awarded after 3-4 years of study (180-240 ECTS).

The Master's degree is awarded after 120 ECTS. The programmes are research-based and are offered in all scientific fields. The Danish title is abbreviated to Cand.[latin abbreviation of academic area] i [field of study]. The English title is Master of Arts (MA) in [field of study] or Master of Science (MSc) in [field of study].

The Master's degree (within fine arts) is awarded after 120-180 ECTS. The programmes are based on research and artistic research. The Danish title is abbreviated to Cand.[latin abbreviation of academic area] [field of study]. The English title is Master of Arts (MA) in [field of study], Master of Music (MMus) [field of study] or Master of Fine Arts (MFA) in [field of study]. Music Academies offer a specialist degree of 2 to 4 years following the master's degree.

The PhD degree is awarded after 180 ECTS. PhD programmes are offered by the universities and some university level institutions offering degrees in the artistic and cultural field.

Detailed descriptions of degree levels can be found in the Danish Qualifications Framework at www.nqf.dk. Please consult the relevant Diploma Supplement for information about the learning outcome of any specific degree.

Adult and continuing higher education

- The programmes normally consist of 2 years of part-time study, equivalent to 1 year of full-time study (60 ECTS credits). Certain master programmes require 1½ years of full-time study (90 ECTS credits). Admission requirements are a relevant educational qualification and at least 2 years of relevant work experience.
- Adult and continuing education is available at levels corresponding to qualifications of the ordinary higher education system.
- The Further Adult Education degree (videregående voksenuddannelse/akademiuddannelse) is awarded after studies at short cycle level and gives access to diploma programmes.
- The Diploma degree (diplomuddannelse) is awarded after studies at first cycle level and gives access to master programmes.
- The Master degree (masteruddannelse) is awarded after studies at second cycle level.

The 7-point grading scale

The grading system used in all state-regulated education programmes as of September 2007 is the 7-point grading scale. Apart from the 7-point grading scale, pass/fail assessment may also be used. 02 is the minimum grade for passing an exam.

Description of grades: 12: For an excellent performance displaying a high level of command of all aspects of the relevant material, with no or only a few minor weaknesses; 10: For a very good performance displaying a high level of command of most aspects of the relevant material, with only minor weaknesses; 7: For a good performance displaying good command of the relevant material but also some weaknesses; 4: For a fair performance displaying some command of the relevant material but also some major weaknesses; 02 For a performance meeting only the minimum requirements for acceptance; 00: For a performance which does not meet the minimum requirements for acceptance; -3 For: a performance which is unacceptable in all respects.